CredVault

Investor Funding Proposal and Production Launch Brief

14 October 2026

Funding at a glance

Funding request	Current maturity	Roadmap
INR 80,000	Sepolia prototype	Six months

Executive summary

CredVault is an existing prototype for verifiable academic credentials. It combines an Ethereum smart contract for credential status and revocation, browser-side document hashing, browser-side AES-GCM encryption, IPFS storage through a serverless Pinata proxy, and wallet-based institution, student and verifier workflows.

The prototype is not yet a production credential service. It is deployed on Sepolia, issuer administration is controlled by a single admin role, Pinata is the effective write dependency, key recovery is limited, and no independent security or privacy review is evidenced. The live external URLs were not independently fetchable in the audit environment.

The proposed INR 80,000 funds production hardening, a scoped independent security review, mainnet deployment and initial ETH reserve, infrastructure and storage reliability, end-to-end QA, institutional documentation and a controlled pilot. It funds validation of an existing technical foundation rather than a new application build.

What has already been built

Area	Evidence-based status
Application	React/Vite app with landing, institution, student, verifier, wallet, proof, QR sharing and key backup workflows.
Smart contract	Solidity 0.8.30 contract with issuer registry, batch issuance up to 100, holder consent, rejection, revocation, reinstatement, expiry, pause and public verification.
Cryptography	Browser `keccak256` document hashing; AES-256-GCM payload encryption; HKDF key derivation; PBKDF2-protected key backups.
Decentralized storage	Encrypted payloads via a serverless Pinata pinning proxy and multiple public IPFS read gateways.
Deployment	Sepolia address `0x26Eb4c3f71ab6735e6c4b5a04D88fa902c46C8B3`; documented deployment block `11,398,037`; broadcast artifacts are checked in.
Testing	Repository documents 56 Foundry tests, 100% coverage claims and 18 frontend crypto tests; fresh local reproduction was blocked by missing npm tools and a Foundry/macOS runner panic.

The target production workflow

1. An authorized institution hashes and encrypts a credential in the browser.
2. Encrypted content is pinned to IPFS; its hash and CID are recorded on the contract.
3. The student accepts the pending credential and keeps a local key backup.
4. The student shares proof or a QR/key payload with a verifier.
5. The verifier reads issuer, status, expiry and document integrity from the chain and encrypted storage.
6. The issuing institution can revoke or reinstate the credential under contract rules.

The core path exists on Sepolia today. Production work must prove it under real network conditions, with stronger administration, independent review, storage recovery, monitoring, privacy documents and controlled institutional onboarding.

Production requirements

Requirement	Existing foundation	Funding enables
Network and gas	Sepolia contract and deployment script.	Reviewed production deployment, verified source, gas benchmarks and capped ETH reserve.
Security and privacy	Threat model, contract tests and Web Crypto; no independent review evidenced.	Scoped independent review, remediation, key/recovery policy and privacy documentation.
Hosting and API	Vercel config, CSP/HSTS, edge pinning endpoint, public RPC/IPFS fallbacks.	Origin/rate controls, quotas, logs, health checks, monitoring and rollback runbook.
Storage reliability	Pinata proxy and gateway failover.	Paid retention, secondary pin/re-pin drill and recovery procedure.
Institutional operations	Issuer registry and authorization instructions.	Onboarding checklist, consent/dispute policy, pilot support and legal/privacy drafts.

The initial release does not require a conventional database. Blockchain state and events remain the credential source of truth. A small operational database can be added later for institution applications, support records and pilot analytics.

Six-month roadmap

Phase	Main deliverable	Allocation
1. Audit and plan, weeks 1-3	Re-run CI, inventory configuration, define network, roles, acceptance tests and release backlog.	INR 8,000
2. Security, weeks 2-7	Independent Solidity, crypto, API and dependency review; remediate critical/high findings.	INR 15,000
3. Mainnet preparation, weeks 5-10	Deployment rehearsal, wallet/multisig, RPC/IPFS/domain/monitoring configuration.	INR 18,000
4. Production, weeks 9-14	Deploy and verify contract, register pilot issuer, publish app, test reliability.	INR 13,000
5. Pilot, weeks 13-20	Proposed 1-2 institution/department pilot, 25 controlled credentials and verification tests.	INR 16,000
6. Evaluation, weeks 18-26	Measure response time, storage recovery, transaction cost, support burden and next-stage needs.	INR 10,000
Total		**INR 80,000**

Phases can overlap. Mainnet launch remains gated by security approval, reliable infrastructure and sufficient ETH.

Funding request and detailed budget

Expense	Work covered	Cost (INR)	Type
Production engineering	Release hardening, admin controls, transaction/error handling, API controls, runbooks and gas benchmarks.	16,000	One-time
Independent security assessment	Scoped review and remediation allowance; not a full formal audit.	14,000	One-time
Mainnet / ETH reserve	Deployment, registration and initial pilot actions; actual gas is variable.	20,000	Variable
Infrastructure	Domain, hosting, RPC, monitoring and six months of basic service.	6,000	Six months
IPFS redundancy	Pinning quota, retention, secondary pin and recovery allowance.	5,000	Six months
Testing and pilot	Browser/device matrix, end-to-end tests, onboarding and support.	9,000	One-time
Legal/privacy documents	Privacy, terms, consent, issuer authorization and dispute/recovery procedures.	4,000	One-time
Pilot maintenance	Bounded updates, incident triage and documentation.	3,000	Six months
Contingency	Fee spikes, provider quota or small launch fixes.	3,000	Reserve
Total funding requested		**80,000**	

One-time/initial launch work is INR 63,000. The six-month operating and reserve portion is INR 17,000. These are portions of the same request.

Operating cost after the funded period

Estimated fixed cost: **INR 2,500 per month**, excluding variable blockchain transactions and taxes.

Monthly item	Estimate
Hosting / edge API	INR 400
RPC provider	INR 700
IPFS pinning / redundancy	INR 800
Monitoring / domain reserve	INR 200
Maintenance / support	INR 400
Fixed monthly total	**INR 2,500**

Gas is calculated separately: `INR = gas units x gas price in gwei x 10^-9 x ETH/INR rate`. For illustration, 3.6 million gas at 12 gwei and INR 250,000/ETH is about INR 10,800; at 30 gwei it is about INR 27,000. These are planning examples, not current quotes.

Funded outcome, risks and impact

Proposed measurable outcomes

- One production contract deployment with verified source and documented admin key procedure.
- Independent scoped security review completed with no unresolved critical finding.
- Complete production issue, accept, share, verify, revoke, reinstate and expiry workflow.
- Proposed pilot of 1-2 institutions or departments, subject to authorization, and at least 25 controlled credentials.
- At least 10 independent verification runs; target median response under 5 seconds under defined healthy-network conditions.
- Storage recovery drill, documented availability baseline, and measured monthly cost per credential.

Key limitations

The admin is trusted. The issuer can regenerate keys under the current deterministic signature model. Shared keys can be copied. Wallet loss can block access. IPFS availability depends on pinning. Mainnet fees fluctuate. Public timing and address relationships remain visible. Blockchain records do not automatically have legal recognition or replace government-issued credentials.

Three launch blockers

1. Independent security and privacy approval.
2. Production network, ETH budget and admin governance.
3. IPFS/API abuse controls and recovery evidence.

Funding justification

CredVault already has the core application and smart contract. The next funding stage is needed because real institutions introduce real security, privacy, operational and economic obligations. INR 80,000 purchases production evidence and controls: review, deployment, gas, resilient infrastructure, QA, documentation and a constrained pilot. Scale decisions should follow measured reliability, per-credential economics and institutional feedback. Before funds are committed, confirm current gas and ETH/INR pricing, the independent security-review quote, hosting/RPC/Pinata/monitoring plans, and legal/privacy fees.